1) $R_{zyz} = R_{z\phi} R_{y\theta} R_{z\psi}$

$$= \begin{bmatrix} c\phi & -s\phi & 0 \\ s\phi & c\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c\theta & 0 & s\theta \\ 0 & 1 & 0 \\ -s\theta & 0 & c\theta \end{bmatrix} \begin{bmatrix} c\psi & -s\psi & 0 \\ s\psi & c\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} c\phi c\theta & -s\phi & c\phi s\theta \\ s\phi c\theta & c\phi & s\phi s\theta \\ -s\theta & 0 & c\theta \end{bmatrix} \begin{bmatrix} c\psi & -s\psi & 0 \\ s\psi & c\psi & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} c\phi(c\theta c\psi + (-s\phi s\psi) & c\phi(c\theta(-s\psi) + (-s\phi c\psi) & c\phi s\theta \\ s\phi c\theta c\psi + c\phi s\psi & -s\phi c\theta s\psi + c\phi c\psi & s\phi s\theta \\ -s\theta c\psi & s\theta s\psi & c\theta \end{bmatrix}$$

2) $R = R_{z\phi} R_{y\theta} R_{x\psi}$

$$= \begin{bmatrix} c\phi & s\phi & 0 \\ s\phi & c\phi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} c\theta & 0 & s\theta \\ 0 & 1 & 0 \\ -s\theta & 0 & c\theta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\psi & -s\psi \\ 0 & s\psi & c\psi \end{bmatrix}$$

$$= \begin{bmatrix} c\phi c\theta & -s\phi & c\phi s\theta \\ s\phi c\theta & c\phi & s\phi s\theta \\ -s\theta & 0 & c\theta \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & c\psi & -s\psi \\ 0 & s\psi & c\psi \end{bmatrix}$$

$$= \begin{bmatrix} c\phi c\theta & -s\phi c\psi + c\phi s\theta s\psi & s\phi s\psi + c\phi s\theta c\psi \\ s\phi c\theta & c\phi c\psi + s\phi c\theta s\psi & -c\phi s\psi + s\phi s\theta c\psi \\ -s\theta & c\theta s\psi & c\theta c\psi \end{bmatrix}$$

3) $H_1^2 = (H_2^W)^T \cdot H_1^W = H_W^2 \cdot H_1^W$

$$= \begin{bmatrix} 0 & 0 & -1 & 4 \\ -0.3827 & 0.9239 & 0 & 1.0824 \\ 0.9239 & 0.3827 & 0 & 2.6132 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 0 & -1 & 0 & -13 \\ 1 & 0 & 0 & 14 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 0 & -1 & 3 \\ 0.9238 & 0.3827 & 0 & 18.99 \\ -0.3827 & 0.9239 & 0 & -4.38 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$H_b^1 = (H_1^W)^T \cdot H_b^W = H_W^1 H_b^W$

$$= \begin{bmatrix} 0 & 1 & 0 & -13 \\ -1 & 0 & 0 & -14 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 & -12 \\ 1 & 0 & 0 & -13 \\ 0 & 0 & -1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$H_b^2 = (H_2^W)^T \cdot H_b^W = H_W^2 \cdot H_b^W$

$$= \begin{bmatrix} 0 & 0 & -1 & 4 \\ -0.3827 & 0.9239 & 0 & 1.0824 \\ 0.9239 & 0.3827 & 0 & 2.6132 \\ 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & -1 & 1 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$= \begin{bmatrix} 0 & 0 & 1 & 3 \\ 0.3827 & 0.3827 & 0 & 2.9302 \\ -0.9239 & 0.9239 & 0 & 3.3786 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

4) $R_x = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\theta & -\sin\theta \\ 0 & \sin\theta & \cos\theta \end{bmatrix}$ $R_x - \lambda I = \begin{bmatrix} 1-\lambda & 0 & 0 \\ 0 & \cos\theta-\lambda & -\sin\theta \\ 0 & \sin\theta & \cos\theta-\lambda \end{bmatrix}$

$\det(R_x - \lambda I) = (1-\lambda)\begin{vmatrix} \cos\theta-\lambda & -\sin\theta \\ \sin\theta & \cos-\lambda \end{vmatrix} = (1-\lambda)((\cos\theta-\lambda)^2 + \sin^2\theta) = 0$

$(\cos\theta-\lambda)^2 = -\sin^2\theta$

$\cos\theta-\lambda = \pm\sqrt{-\sin^2\theta}$

if $j=\sqrt{-1}$; $\sqrt{-\sin^2\theta} = \sqrt{-1}\cdot\sqrt{\sin^2\theta} = j\cdot\sin^2\theta$

$\cos\theta-\lambda = \pm j\cdot\sin\theta$

$\lambda = \cos\theta \pm j\sin\theta$

$\lambda_2 = \cos + j\sin\theta$, $\lambda_2 = e^{j\theta}$

$\lambda_3 = \cos - j\sin\theta$, $\lambda_3 = e^{-j\theta}$

5) if $\lambda=1$, $RV=V$

That means when the rotation matrix is applied to the specific vector, the vector's direction and length remain unchanged.